VOLLEY: A Linear-Motor Electromagnetic Deployment System for Deterministic CubeSat Orbit Seeding from Small Launch Vehicles

Adityavardhan Mishra
Department of Mechanical Engineering, Symbiosis Institute of Technology
Symbiosis International (Deemed University), Pune, India
PRN 23070125054

***Abstract*—Secondary payloads on rideshare missions inherit the orbit of the primary customer, and the spring deployers that release them impart 1–2 m/s, too little to alter that orbit in any useful way. This paper presents the design and analysis of VOLLEY, a magazine-fed electromagnetic deployer that ejects unmodified CubeSats at a programmable velocity an order of magnitude above spring systems while remaining within standard qualification loads. On-orbit electromagnetic launch of CubeSats has been studied before, at velocities twenty times higher but at accelerations of order 10^3 g and with a conductive armature attached to the payload [27]; the contribution claimed here is not electromagnetic deployment as such but the combination of a programmable velocity with a satellite that is never modified and an acceleration inside its existing qualification envelope. An architecture trade selects an ironless double-sided Halbach-array linear synchronous motor driving a reusable permanent-magnet sled along a 1.5 m track, with twelve 3U satellites fed from two transverse cassettes, a contactless eddy-current arrest brake, and supercapacitor pulse power. A winding-resolved motor model, cross-checked against an independent magnetostatic computation, gives a thrust constant of 10.54 N per kA/m with $\pm$1.0% ripple and an exit velocity of 16.0 m/s for a 3U payload at 10.1 g, drawing 2.78 kJ per shot. Regenerative braking of the reusable sled over 39 mm of stator downstream of release returns 0.05 kJ of the sled's 1.27 kJ to the bank at the same sheet-current rating, giving 18.8% electricaltopayload efficiency net of recovery; the eddy brake remains necessary and still absorbs 1.16 kJ per shot. The velocity follows from a 9.4 kg sled mass computed from the detailed CAD solid volumes; an earlier parametric estimate of 4.9 kg gave 20.4 m/s, and the decision to adopt the CAD-derived value was fixed in advance of the structural analysis that tested it. Closed-loop simulation with realistic sensing yields a velocity dispersion of 0.0274 m/s (3σ) about a 15.8 m/s fleet setpoint, equivalent to $\pm$0.10 km of apogee placement. One maximum-velocity ejection multiplies a propulsion-less satellite's orbital lifetime by 1.60—7.3 times the extension a 2.5 m/s spring delivers, since lifetime extension is superlinear in Δv and the raw velocity ratio of 6.4 understates it; that ratio compares *gains*, and on delivered orbital life it is 1.50, which is the figure any risk-weighted comparison must use because a satellite the deployer fails to release delivers nothing; an independent propagator reproduced the earlier 20.4 m/s operating point at mean and high solar activity but not at low; the current 1.60 multiplier has not been independently rerun, so the ratio is quoted at a stated activity level and is not claimed invariant, and one ejection raises semi-major axis by 28.8 km, which is the property no alternative reaches: satellites released at different times from the same host separate in true anomaly within a single orbit at no velocity cost, so 30° of in-track phase costs 468 s of waiting and is not claimed here, while release timing changes semi-major axis by 0 m at any cadence. The mass model closes the system at 126.6 kg dry and 174.6 kg loaded; its component lumps are not physical measurements. Host integration is developed generically for any restartable upper stage or hosted platform, with ISRO's POEM and Skyroot's Vikram-1 kick stage analyzed as candidate first hosts. The concept of operations treats the host as an active participant rather than a mounting surface: after primary separation a spent stage repositions between altitude shells on its own reaction control and fires satellites at each station before disposal, which a reachable-envelope analysis prices at 27.8 m/s per 50 km shell and bounds by excluding plane change entirely at 133 m/s per degree.**

***Index Terms*—CubeSat deployment, electromagnetic launch, Halbach array, linear synchronous motor, rideshare, constellation phasing, small launch vehicles.**

I. Introduction

The rideshare model has cut the cost of reaching orbit for small satellites, and it has done so by removing their choice of destination. A CubeSat manifested as a secondary payload is released wherever the primary customer's mission ends, through a spring deployer whose 1–2 m/s separation velocity exists for clearance rather than for orbit shaping. Satellites that carry propulsion can correct for this. The large class that cannot, which includes most university missions and cost-constrained commercial payloads, remains wherever it was dropped, at whatever altitude and phasing the manifest happened to produce. That class is the majority: of the more than 4,800 nanosatellites and CubeSats catalogued as of January 2026, on the order of 222 carry a propulsion system [24]. The population is also still growing, with roughly 16,900 satellites under 500 kg forecast for 2026–2035, about a third of all satellites launched but only some 6% of launched mass [25]—so the constellation, whose members must be *distributed* before it functions as one, is now the dominant architecture by count while the deployment interface that separates them has not changed in twenty years.

Two families of hardware bracket the problem. Below it sit flight-proven springs, typified by the J-SSOD at 1.1–1.7 m/s [2] and commercial deployers near 2 m/s [20]. Above it sit propulsive orbital transfer vehicles, which move an entire bus through hundreds of m/s at a cost, mass, and integration burden far beyond what a 4 kg satellite needing a 20 m/s nudge can justify. Between the two lies a regime unserved *by flown* *hardware*: no deployment system in service operates in the tens of m/s, although the astrodynamic utility of that band is substantial and quantifiable. The band is not unexamined in the literature. Feng *et al.* simulate an on-orbit three-stage induction coilgun accelerating a 20 kg CubeSat to 321.56 m/s and map the resulting reachable domain [27], and a sustained line of work from a second group develops magazine-fed electromagnetic transfer and release of stacked CubeSats, carried as far as a measured prototype of the transport mechanism [30], [28], [29]. The contribution claimed here is therefore narrower and specific: a programmable exit velocity delivered to an *unmodified* satellite within its own qualification envelope. Both qualifiers do work. An induction coilgun accelerates a conductive armature, which must either be attached to the customer satellite or separated as a sabot. And the velocity such a machine makes available is bounded in practice not by the machine but by the payload: Feng's 321.56 m/s is reached over a 3.9 m barrel, a mean acceleration of 1.35×10^3 g, with a reported peak force above 600 kN on a 20 kg body—some 3×10^3 g—against the 14 g quasi-static qualification level a standard CubeSat is built to [21]. That gap, not the achievable speed, is what confines useful exit velocity to the tens of m/s, and it is the constraint this design was built around.

This paper develops a system for that band and keeps the treatment deliberately host-agnostic: the deployer is specified against a generic interface that any restartable upper stage, kick stage, or hosted orbital platform can satisfy, and specific vehicles enter only as worked candidate examples. The contributions are (1) an architecture trade showing why a linear synchronous motor rather than a coilgun is the correct electromagnetic topology at CubeSat structural limits; (2) a complete system design covering magazine, restraint, arrest, power, and abort; (3) a winding-resolved electromagnetic model, independently verified, together with ten quantitative analyses spanning launch performance, astrodynamics, deployment safety, and host interaction; and (4) a generalized integration framework with capacity and recoil budgets evaluated for candidate hosts in the Indian launch ecosystem.

II. Related Work

A. Deployment heritage

The P-POD established the rail-and-spring standard, and the ISS-based J-SSOD and NanoRacks systems industrialized it [2], [3]. The failure record is instructive: the NanoRacks ball-lock anomaly, in which jack-screw preload above 0.11 N·m drove the release mechanism toward seizure, traced to a load path in which ascent preload passed through the release device itself [1]. VOLLEY's restraint architecture is designed against precisely this fault class. The closest electromagnetic relative is a proposed multi-CubeSat deployer using permanent-magnet conveying and ejection stages, demonstrated at 0.4–1.4 m/s [4]; VOLLEY extends the same Lorentz-force principle to roughly fifteen times that velocity.

B. Electromagnetic launch and supporting technology

Coilgun research examined nanosatellite launch at km/s scales [5], [6], establishing contactless acceleration and its costs: the single-stage reluctance machines of that lineage convert 1–2% of stored energy into kinetic energy, though multi-stage induction designs do far better, and the on-orbit CubeSat launcher of [27] reports 14.9–19.9%. Ironless permanent-magnet linear machines occupy the opposite corner of the design space: the Halbach array concentrates flux on one working face [7], and the Inductrack program demonstrated large forces from such arrays with no iron and no sliding contact [8]. Space-qualified supercapacitors have progressed to an established supply chain through ESA-led programmes [9], [10]. Eddy-current dampers carry decades of flight heritage in deployment mechanisms [11]. On the astrodynamics side, Planet Labs demonstrated propulsion-free constellation phasing by differential drag over campaigns of weeks to months [12], [13]; that timescale is the benchmark against which impulsive seeding is measured. ISRO's POEM programme converted the spent PSLV fourth stage into a stabilized hosted-payload platform across four missions, closing POEM-3 with a controlled, debris-free reentry [14], [15], establishing the regulatory template for operating equipment on a spent stage.

III. System Architecture

A. Requirements

The driving requirements are: eject an unmodified 3U CubeSat, imposing no armature, plating, or electrical interface on the customer satellite; keep payload acceleration within standard qualification loads (25–30 g); mount on a restartable upper stage, kick stage, or hosted orbital platform through a standard ESPA bolt pattern, a host class whose accommodation envelope is not public and against which the closed 1839 mm length therefore cannot yet be shown to fit (Sec. XV); provide per-satellite programmable velocity with dispersion small against its astrodynamic effect; carry a twelve-satellite manifest with autonomous sequencing; provide inhibits and a defined abort; and hand recoil to the host in a form its attitude control absorbs at negligible propellant cost.

B. Derivation of Principal Design Parameters

The values that define the machine are set by requirements and standards rather than chosen freely; Table I records the governing relation for each. Three deserve explicit statement. The acceleration ceiling follows from the CubeSat Design Specification qualification environment: NASA GEVS [21] protoflight random vibration integrates to 14.1 $g_{\rm rms}$, whose 3σ envelope of roughly 42 g bounds the launch load, and a 25 g design cap preserves margin against it. An earlier version of this design was acceleration-limited: at a 4.9 kg parametric sled the ejection ran at 16.3 g, close enough to the cap that the cap set the velocity through $v_{\max} = \sqrt{2a_{\max}L}$. That is no longer the binding constraint. With the sled computed at 9.4 kg from the CAD solid volumes the same commanded force accelerates a heavier moving mass, so the ejection sits at 10.5 g and the 1.3 m zone yields 16.0 m/s. The machine is

TABLE I
GOVERNING RELATION FOR EACH PRINCIPAL PARAMETER

Parameter	Value	Set by
Accel. cap	25 g	GEVS 14.1 g_{rms}, 3σ [21]
Track length	1.5 m	$v = \sqrt{2aL}$ + coast zone
Rail material	7075-T6, hard anodize	CDS rail spec [23]
Bank voltage	96 V	SiC 1200 V derated 60%
Bank capacitance	6 F	droop eqn (2), $\leq$5% SoC sag
Sled magnets	N45SH	thermal $\times$ field (Sec. V-F)
Brake cap	200 g sled	magnet bond MoS (Sec. III-F)
Abort point	45% stroke	braking-distance equality
Magazine	12 (2$\times$6)	cassette CoM symmetry

TABLE II
TOPOLOGY TRADE AT CUBESAT STRUCTURAL LIMITS

Criterion	Reluctance coilgun	Ironless LSM + sled
Velocity ceiling (payload-limited)	26–35 m/s	26–35 m/s
Electrical$\rightarrow$payload efficiency	14.9–19.9% [27]	18.8% (Sec. V)
Satellite modification	armature required	none
Velocity control	pulse timing	closed-loop servo
Abort	none once fired	to $\sim$45% stroke
Field on stored satellites	pulsed, unshielded	Halbach self-shielded

now *thrust-and-mass* limited rather than acceleration-limited, with more than a factor of two of qualification margin left unused. Recovering velocity therefore means removing sled mass or raising sheet current, not shortening the stroke; a 0.2 m coast-trim zone completes the 1.5 m axis. Stroke buys velocity only as $\sqrt{L}$, so the trade is unattractive: total track lengths of 1.0, 1.2, 1.5 and 1.8 m give 12.9, 14.4, 16.4 and 18.2 m/s respectively, and the envelope is already the binding packaging constraint (Sec. XV). The capacitor bank follows from the shot energy and the droop it actually reaches: at the 2.78 kJ draw the selected 6 F bank sags to $V_1 = 0.947V_0$, and

$$C = \frac{2E}{V_0^2 - V_1^2} = 6.00 \text{ F} \rightarrow 6 \text{ F}, \tag{1}$$

realized as thirty-two series 190 F, 3 V commercial cells for a 96 V rail compatible with 1200 V SiC devices at a comfortable derating.

C. Topology Trade

The payload settles the trade before the electromagnetics do. Exit velocity is bounded by

$$v_{\max} = \sqrt{2\,a_{\max}L}, \tag{2}$$

so at 25–30 g over a 1.3–2 m stroke the ceiling is 26–35 m/s. The coilgun's defining advantage, a km/s-class ceiling, is unreachable by the payload it would carry, while its costs remain: microsecond timing against the suck-back effect, an armature bolted to the customer's satellite, and no abort once fired. Efficiency is deliberately not among them: at 14.9–19.9% [27] a multi-stage coilgun is comparable to the figure reported here, and an earlier version of this argument claimed otherwise on the strength of a single-stage number. A linear synchronous motor concedes nothing in the reachable envelope and inverts every cost: high drive efficiency, continuous servo control, and a reusable sled that carries the magnets so the satellite carries nothing. Table II summarizes.

D. Layout and Firing Cycle

The launcher is organized around a single 1.5 m track: a 1.3 m acceleration zone and a 0.2 m coast-and-trim zone (Figs. 1 and 2). The stator is an ironless double-sided three-phase copper winding; the mover is a 9.445 kg sled carrying opposed Halbach arrays (48 mm wavelength, 8 mm

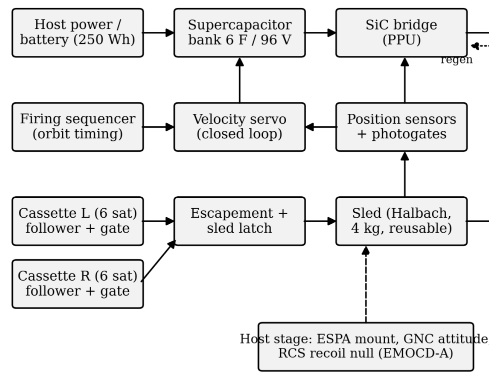

Fig. 1. System block diagram: power, control, mechanism, and host-interface chains.

N45SH blocks, four blocks per wavelength) across a 12 mm winding gap. Two transverse cassettes of six 3U satellites flank the breech. The cycle: a follower advances one pitch and slides a satellite laterally onto the sled cradle (a sub-newton operation in microgravity); a detent latch engages; a two-finger escapement retains the next satellite; the motor accelerates the stack; force is removed in the coast-trim zone where the servo makes its final correction; the sled enters the brake and decelerates while the satellite departs along guide extensions. Retention during acceleration costs nothing, since inertia presses the satellite into the aft backstop, and release costs nothing, since braking the sled is the release. Internal mass motion produces transient counter-rotation, not a residual rigid-body rate after the mass stops. The corrected A13 budget gives a peak return rate of 0.130 deg/s for the assumed 500 kg host and 0.385 deg/s at 200 kg, so its two declared peak-rate bands remain failed. The ideal rigid-body residual is zero, and the previous 8.2 s rate-null and 18.1 s cadence floor are superseded. The motion instead leaves an attitude offset—0.40 deg in the 500 kg example under the assumed 166 mm lever arm—whose restoration time depends on a controller and schedule not modelled here. Structural ringing is also unmodelled. The 10–20 s and 1200 s inter-shot figures the analysis previously carried in parallel are reconciled: **1200 s is the adopted ConOps cadence**, with 10–20 s describing the machine's recharge floor rather than its operating point, so twelve shots span 4.0 h and about 2.6 orbits.

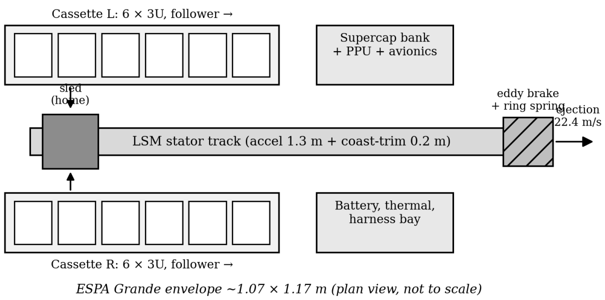

Fig. 2. Plan-view layout against an ESPA-Grande-class envelope (not to scale), retained as a dimensional reference. The closed envelope exceeds that class in length by 44%; ESPA-Grande port compliance is not a requirement of this design, and the target host class is discussed in Sec. XV.

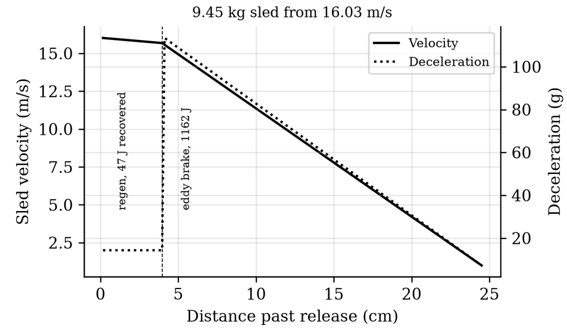

Fig. 3. Eddy-brake arrest, taper-limited to 200 g on the sled.

E. Restraint, Inhibits, and Abort

Launch restraint separates the preload path from the release path, the inverse of the NanoRacks fault [1]. One-shot retention gates at each cassette exit carry ascent preload directly into structure; the escapement is caged during ascent; the sled is held by a motorized over-center cam lock. Firing requires a three-inhibit chain: redundant seat switches, an attitude-valid flag from the host, and a sequencer arm. Abort is available to approximately 45% of the stroke, beyond which braking distance for the combined mass exceeds remaining track; the muzzle buffer is sized to capture an aborted combined mass. The retention gate carries the ascent preload of a six-satellite stack through two D9 A-286 shear pins of 41.0 kN double-shear capacity; the escapement, caged behind the gate, sees none of this. The quasi-static case, $24\,\mathrm{kg} \times 25\,g = 5.9$ kN, is *not* the sizing case, and the gate was originally sized against it with two D6 pins. Miles' equation on the GEVS protoflight spectrum (14.1 g_{rms}, 0.16 g^2/Hz) at the track's 109 Hz fixed-fixed mode gives a 3σ load of 11.7 kN at $Q = 10$ and 20.2 kN at $Q = 30$, two to three and a half times the quasi-static figure, which left the D6 margin negative at $Q = 30$. The pins were not undersized; the load case was. Resizing to D9 raises the margin of safety to $+0.45$ at $Q = 30$ and keeps it positive across $Q = 10$–30 for an added mass of 11 g, so the retention chain no longer depends on where the unmeasured structural damping lands—which matters because the same unmeasured Q governs the force-ripple resonance discussed in Sec. XV and cannot be resized away there. Random vibration, not quasi-static preload, is the driving case for the retention chain, and is now the case it is sized against. The abort threshold is where the sled braking distance under the same thrust equals the remaining track, which for the loaded 13.4 kg carriage falls at 45% of stroke and 11.1 m/s. The commit fraction is unchanged by the heavier sled—braking distance and travelled distance scale with mass identically, so the mass cancels—but the velocity at the commit point falls with the acceleration.

F. Arrest, Power, and Materials

The sled leaves release with about 1.27 kJ. Motor regeneration alone cannot arrest it, because braking force is bounded by the same thrust constant as acceleration; that bound sets what regeneration can and cannot do, and both halves matter. Braking at the rated sheet current over the 39 mm of stator the closed envelope allows downstream of release extracts 334 J of mechanical work for 15 J of copper, returning **291 J** to the bank, and the sled still reaches the brake at 14.1 m/s carrying **935 J**. A copper-fin eddy-current brake therefore provides bulk arrest: force proportional to velocity, contactless, with flight heritage in the damper class [11]. Authority is abundant, so the pole entry is tapered to cap sled deceleration near 200 g. This limit is set by the magnet retention: a 65 g Halbach block at 200 g loads its bonded interface (10.8 cm^2) to only 0.12 MPa against a conservative 10 MPa epoxy allowable, a margin of 84, so 200 g is a deliberate conservatism protecting the brittle sinter rather than a strength limit. The 0.344 kg copper fin absorbs the sled's residual 935 J per shot at an adiabatic 7.1 K per-shot transient rise; cooling between shots has not been modelled; a ring-spring catches the residual below 1.5 m/s (Fig. 3). Pulse energy comes from a 6 F, 96 V supercapacitor bank switched by a SiC bridge [9]. Materials follow three rules: nothing conductive rides the changing field (titanium sled chassis, PEEK formers), nothing near the track is ferromagnetic except by design (BeCu springs, A-286 fasteners), and nothing outgasses near customer optics (E595-compliant polymers, dry films, sealed roller cartridges).

IV. Models and Verification

A. Field Model and Independent Verification

The Halbach airgap field is a decaying wave, $B(y) = B_0 e^{-ky}$ with $k = 2\pi/\lambda$, summed over the two opposed faces. Because this carries the force budget, it was checked against an independent magnetostatic computation (analytic cuboid superposition, magpylib), with the array rotation-sense convention determined empirically by probing a single array on both faces; the exercise caught two successive sign errors whose outputs matched single-array fingerprints. Converged results: single-array mid-gap 0.351 T against the wave model's

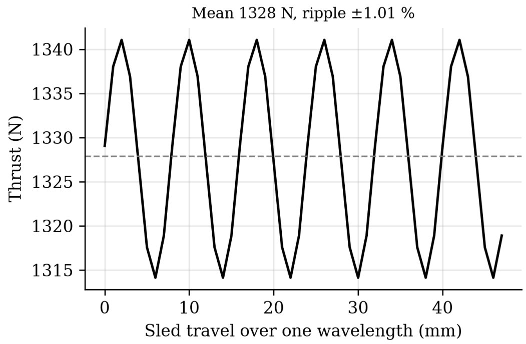

Fig. 4. Winding-resolved thrust over one wavelength of sled travel.

0.351 T; double-sided mid-gap peak 0.694 T against a predicted 0.702 T. The stray field behind the array back face, which sets the magnetic keep-out for stored satellites, computes to 22.7 mT at 10 mm, 4.3 mT at 20 mm, and 0.4 mT at 50 mm.

B. Winding-Resolved Thrust Constant

The airgap field is sampled in all three dimensions before the Lorentz sum: B_y is Gauss-Legendre averaged across the array's full 90 mm depth rather than taken on the centre plane. That distinction is worth stating because the centre-plane value is 4.4% higher, and using it—as this work did until an independent depth-resolved integral was run—overstates the thrust constant by the same margin and the exit velocity by 2.2%. A slotless three-phase belt winding (pole pitch $\lambda/2$, belts of $\lambda/6$, winding factor 0.955) was resolved against the verified field by direct Lorentz integration with position-locked sinusoidal commutation, the field translated with the sled. The result is a thrust constant

$$K_t = 10.54\,\mathrm{N\,per\,kA/m}, \qquad \text{ripple } \pm 1.01\% \text{ (6th harmonic)}, \tag{3}$$

shown in Fig. 4. Synchronous extraction carries the one-half traveling-wave factor that lumped $\langle B \rangle KA$ models hide; the corrected constant is recovered operationally by rating the sheet current at 140 kA/m and commanding 126 kA/m (21 A/mm^2 pulse-only in the copper), where adiabatic heating remains 0.3 K per 162 ms shot.

C. Shot, Servo, and Astrodynamic Models

The shot integrates sled-plus-satellite dynamics against the supercapacitor circuit with explicit copper loss and a 95% converter. Velocity control uses a position-scheduled profile with proportional correction and a coast-trim residual correction from photogate measurement; an 800-run Monte Carlo over thrust-constant, mass, force-ripple, and sensor tolerances gives the dispersion. The proportional gain is not tuned to that dispersion but designed against stability margins, as Sec. IV-D sets out. Orbital lifetime uses per-revolution orbit-averaged decay (Gauss tangential equations, quadrature over eccentric anomaly, static exponential atmosphere at mean activity [17]); the method was cross-validated against an independent Cowell RK4 propagation with drag, the two agreeing on 30-day semi-major-axis decay to 99.4%. Constellation seeding uses $\Delta a = 2a\,\Delta v/v$ and the resulting period difference; deployment safety propagates the stage and all deployed ellipses with Kepler-plus-secular-J_2 motion for 30 days, with candidate minima refined at 0.25 s sampling.

D. Velocity-Loop Stability

The control law divides the commanded sheet current by the modelled thrust constant and multiplies by the modelled mass, so the plant's own K_t/m cancels and the open-loop transfer reduces to $L(s) = K_p\, e^{-s\tau}/s$. The proportional gain is therefore not a current gain but an acceleration per unit velocity error, in s^{-1}, and *its numeric value is the gain crossover frequency in rad/s*. The total lag τ is the sensing and computation transport delay plus the half-sample of a zero-order hold; both are pure delays and add. No sensor has been selected, so the delay is a stated assumption, swept rather than asserted, and 0.6 ms is used as the reference value with a 0.1 ms hold at 5 kHz.

That reduction sets the design constraint. An earlier gain of 3500 s^{-1} places the crossover at 557 Hz—five times the track's 109 Hz first bending mode and above the 48 Hz mode as well—giving $-50.4°$ of phase margin and -3.9 dB of gain margin at the reference delay, with marginal stability reached at only 0.45 ms of total lag (Figs. 5 and 6). Two properties of the simulation had concealed this: the loop fed back an undelayed state, which is the one condition under which that gain is stable, and the command is clipped at the winding rating, so the linearly unstable loop behaves as a bang-bang relay whose mean follows the feedforward term while the terminal photogate trim removes the residual. The reported dispersion was consequently governed by the saturation limits and the terminal correction rather than by the feedback it was attributed to, which is why the closed-loop figure is reported here only alongside the margins that make it a property of a design.

The gain is instead sized as the largest value simultaneously holding $\geq 50°$ of phase margin at the reference delay and keeping closed-loop bandwidth at or below one third of the first bending mode, which gives 195 s^{-1}: crossover 31.1 Hz, phase margin $+82.2°$, gain margin $+21.2$ dB, and no excursion above the current rating anywhere in the stroke. **The gain falls by a factor of eighteen and the 3σ dispersion does not move**, 0.0271 to 0.0267 m/s. The loop never required the bandwidth: dispersion is set by the terminal trim and by the thrust-constant and mass tolerances, not by loop gain. The binding constraint is now the structural one, so a measured first mode below 109 Hz would lower the designed gain with it. Classical margins on a 162 ms trajectory are necessary rather than sufficient, the plant is treated as rigid, and the structural modes enter only as a frequency the bandwidth is held away from; a compliant-track model remains outstanding.

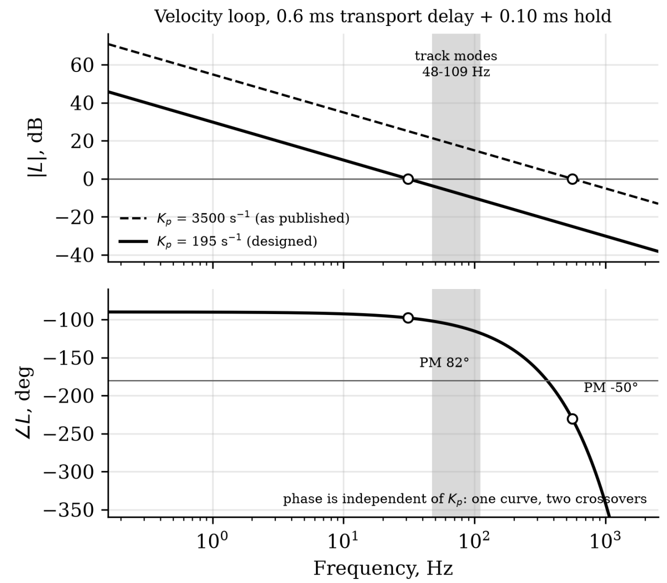

Fig. 5. Open-loop velocity-loop response at the earlier and the designed gain, with the 48–109 Hz band of track modes shaded and both gain crossovers marked. Phase depends only on frequency and delay, so the two gains share one phase curve.

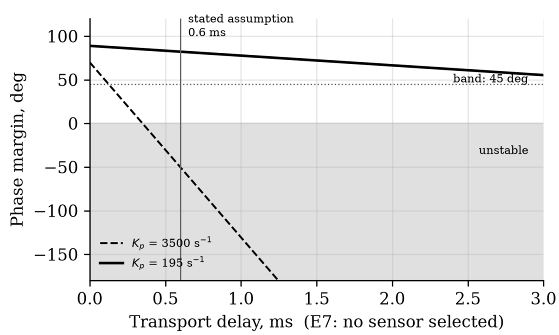

Fig. 6. Phase margin against transport delay. The earlier gain crosses into instability at 0.35 ms of transport delay; the designed gain holds more than 55° across the whole swept range.

V. Performance Results

A. Launch Performance

At the rated point (3U reference, 9.445 kg sled computed from the CAD solid volumes, 1.3 m acceleration): exit velocity **16.0 m/s** at **10.1 g**, pulse 162 ms, peak current 320 A, and bank stateofcharge sag 5.2% (Fig. 7). Energy drawn is 2.78 kJ per shot, with copper heat of 835 J and a further 83 J dissipated in the bank's own series resistance. Regenerative braking returns 291 J after release, so the net draw is 2.74 kJ. The payload receives 514 J, a **18.8% electricaltopayload efficiency**; the sled's share is split between 291 J recovered and 935 J dissipated in the arrest brake. Efficiency falls with the heavier sled for two compounding reasons: a larger fraction of the same mechanical work goes into a mass that is then braked away, and the longer 162 ms pulse accrues more copper loss at unchanged current density. Recovery is what partially offsets

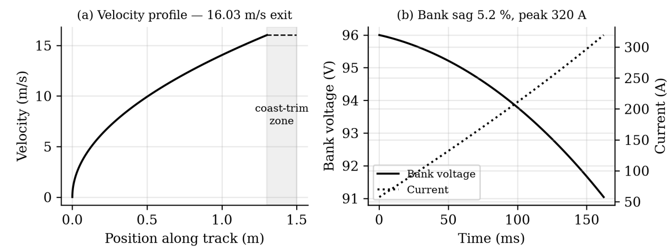

Fig. 7. Shot profile at the rated point: (a) velocity with coast-trim zone; (b) bank voltage and current.

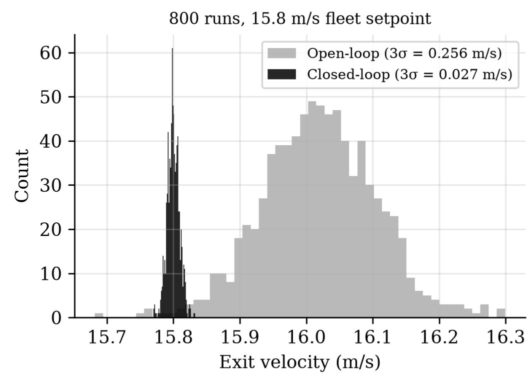

Fig. 8. Closed-loop exit-velocity dispersion, 800 Monte Carlo runs.

the first of those, and it is bounded by geometry rather than by loss: over the available 39 mm the braking pulse lasts 15.8 ms and burns 15 J of copper, against 835 J over the 162 ms shot, so recovered energy rises monotonically with braking force up to the winding's rating and with distance beyond it. A 500 mm section would return 570 J and a 1 m section would arrest the sled outright while returning 783 J, but the closed envelope already exceeds its target class by 44% (Sec. XV), so distance is not available here. Peak current during recovery is 219 A, below the shot's own 320 A, so the drive is not re-rated. The closed-loop Monte Carlo returns a dispersion of **0.0274 m/s (3σ)** at a 15.8 m/s fleet setpoint (Fig. 8), dominated by the sensing chain rather than plant tolerances. The setpoint must sit below the open-loop ceiling for the servo to retain authority; at 98.85% of it, the nominal headroom is 0.188 m/s. Table III gives the payload family: the system is force-limited above 1U, so heavier payloads trade velocity, never structural margin, and the deployer mass each customer pays for varies by a factor of 109 across the family (Sec. V-B).

B. Payload Class and the Mass-per-Satellite Argument

Table III is read most usefully from its last column rather than its first. Exit velocity is nearly flat across the family, because the 9.445 kg sled is most of the moving mass: removing the payload entirely buys 19% of velocity, so velocity is a property of the sled and the stroke and not of the customer. Shortening the magnet array does not recover it either, since array length sets thrust and sled mass together—a 150 mm

TABLE III
PAYLOAD FAMILY AT THE RATED POINT (1.3 M ACCELERATION, 9.445 KG CAD-DERIVED SLED). DEPLOYER MASS PER SATELLITE USES THE 126.6 KG DRY MASS AND A PACKING MODEL CALIBRATED SO THE 3U CASE RETURNS THE TWELVE THE MAGAZINE IS LAID OUT FOR. EVERY COLUMN IS REGENERATED FROM `PAYLOAD_FAMILY.JSON` AT THE CURRENT OPERATING POINT; THE VALUES PUBLISHED UNTIL 2026-08-20 PREDATED BOTH ADR-030 AND THE ENCLOSURE BUILDUP

Class	Exit vel.	Accel.	Per load	kg/sat.
PocketQube 1P (0.25 kg)	18.9 m/s	14.0 g	326	**0.388**
ThinSat (0.28 kg)	18.8 m/s	13.9 g	123	1.029
PocketQube 3P (0.75 kg)	18.4 m/s	13.3 g	108	1.172
TubeSat (0.75 kg)	18.4 m/s	13.3 g	41	3.088
1U (1.33 kg)	17.9 m/s	12.6 g	40†	3.165†
3U (4 kg)	16.0 m/s	10.1 g	12	**10.550**
6U (8 kg)	14.1 m/s	7.8 g	6	21.100
12U (12 kg)	12.7 m/s	6.3 g	3	42.200

array gives a 5.15 kg sled but a 4.95 N per kA/m thrust constant, and the 3U case falls to 13.3 m/s.

The column that does move is deployer mass per satellite, by a factor of thirty. At 3U the machine spends **7.04 kg of deployer per customer**, which loses to a cold-gas propulsion module delivering the same 16.0 m/s at 0.5–1.2 kg by roughly a factor of eight. At PocketQube it spends **0.24 kg** and wins by two to five. The comparison inverts entirely on payload class, and on no machine parameter at all: not the thrust constant, not the stroke, not the sheet current.

Three qualifications belong with those counts rather than beneath them. First, they are volumetric. The packing efficiency is calibrated against the one configuration that has been laid out in CAD—the 3U case, where a raw volume ratio predicts 21 and the magazine holds 12—and the resulting 56% is then applied to classes for which no cassette exists. Second, the feed engages the CubeSat Design Specification corner-rail interface at the cradle, the escapement and the retention gate; PocketQubes, ThinSats and TubeSats do not have that interface, so a smaller class needs its own cassette, cradle and gate, which is mechanical design rather than a parameter change. Third, hundreds of satellites per load is a different machine from twelve: the campaign thermal case, the bank recharge duty and the escapement cycle life were all sized for twelve shots, and the bank is already the binding constraint at that number (Sec. XV). What Table III establishes is that payload class dominates the economic argument, not that a small-payload variant has been designed.

C. Astrodynamic Utility

A prograde ejection raises apogee while leaving perigee at the deployment altitude, so the correct claim is a longer-lived ellipse. At 450 km with ballistic coefficient 61 kg/m^2, one 16.0 m/s shot multiplies lifetime by **1.60** at mean solar activity (base lifetimes 0.52–2.61 yr across the activity range; Figs. 9 and 10).

An earlier version of this work claimed the multiplier was invariant to two decimal places across ballistic coefficients of 40–90 kg/m^2 and across a fivefold density range, and

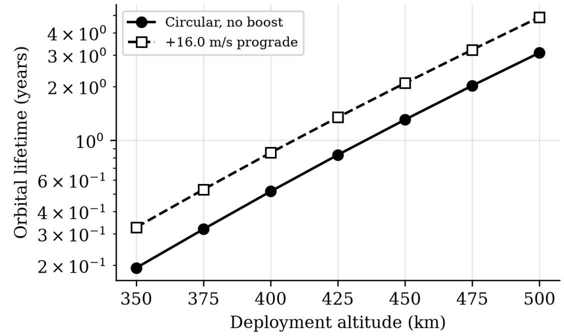

Fig. 9. Orbital lifetime vs deployment altitude (BC = 61 kg/m^2, mean activity).

offered that invariance as the result the analysis could defend. That claim does not survive an independent propagator. Run headless against MSISE90 with 20×20 gravity, lunisolar third bodies and solar radiation pressure, GMAT R2022a reproduces the multiplier at mean and high activity but returns 2.07 at low activity, a spread of 18% against a 5% acceptance band fixed before the run. The reason is a defect in how the sweep was posed rather than in the arithmetic: the model represents solar activity as a uniform multiplicative scale on density, and ballistic coefficient enters the drag term in the same multiplicative slot, so both sweeps divide the two lifetimes being compared by the same factor. Their ratio cannot move, whatever the atmosphere does. A real atmosphere changes the *shape* of the density–altitude profile with activity, the boosted orbit's apogee samples that profile some 37 km higher than the baseline's, and the ratio then does move. The multiplier is therefore quoted here at a stated activity level, and no invariance is claimed.

Differential ejection velocities put adjacent satellites on different periods: splits of 2, 5, and 10 m/s reach 30° spacing in **6.9, 2.8, and 1.4 days**. Under identical density and geometry assumptions, differential drag between 3:1 area configurations achieves the same spacing in **25.0 days** (Fig. 11), consistent with flown drag-phasing campaigns operating over weeks to months [12]. Impulsive seeding followed by drag trim follows naturally as a hybrid.

D. Deployment Safety

Every prograde shot leaves an ellipse whose perigee sits at the firing altitude, so the fleet re-crosses the stage's orbit by construction, and the along-track phase between a deployed satellite and the stage realigns on a period of **9.9 days** at the rated velocity (Fig. 12). The closest approach reached within a 30-day screening window is deliberately *not* quoted here as a safety figure: because both bodies re-cross the same altitude, that minimum is fixed by the exact along-track phase at crossing, which is acutely sensitive to ejection velocity—in the model a $\pm 2.5\%$ change about the rated point sweeps the 30-day minimum by more than an order of magnitude. It is a near-resonant sample of the beat geometry, not a design property. Two statements survive that sensitivity and

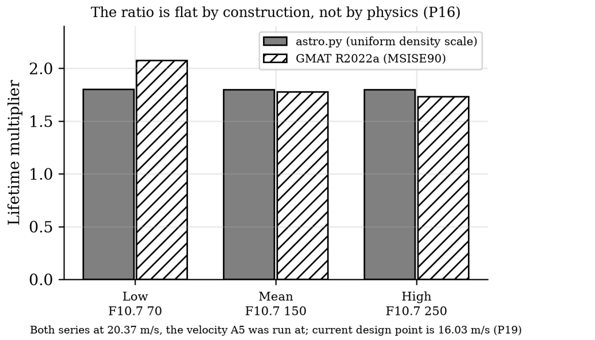

Fig. 10. Solar-activity sweep against GMAT. The static-atmosphere model returns a flat ratio by construction, not by physics; an independent propagator does not.

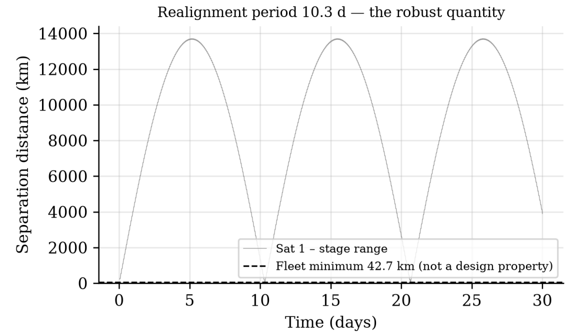

Fig. 12. Satellite–stage range over 30 days, staggered firing at the rated velocity.

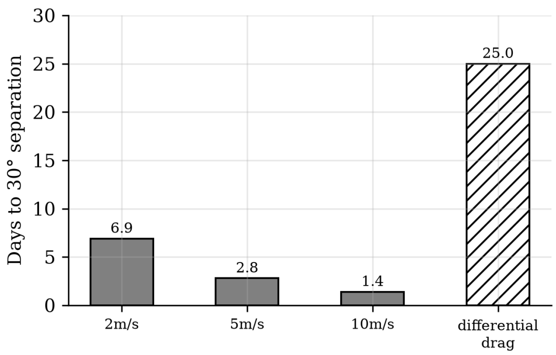

Fig. 11. Constellation seeding: VOLLEY velocity splits versus differential drag under identical assumptions.

carry the safety argument. First, per-shot collision-avoidance (COLA) screening is **mandatory** for every ejection, because the approach geometry cannot be certified from the nominal velocity alone. Second, disposing of the host stage before the first 9.9-day realignment—routine for a restartable stage—removes the recurring close-approach mechanism at its source, before the fleet can return to the stage's along-track position. Per-shot conjunction products remain a mission-operations deliverable.

E. Error, Tip-Off, and Mass

The 0.0274 m/s dispersion maps to **±0.10 km apogee placement** at 450 km: sub-kilometre deterministic insertion for a satellite with no propulsion. Tip-off is modelled rather than bounded, and the risk is not where the analytic budget put it. Release occurs 200 mm—12.2 ms—into the coast-and-trim zone with commanded force already at zero, so at the residual latch and friction load the mechanism tolerates roughly 250 μs of asymmetry and still lands two orders of magnitude inside the 2 deg/s comparator. What is not benign is the start of the stroke: the payload centre of mass sits 70 mm off the thrust line, the 413 N push therefore applies a 28.9 N·m moment, and the payload accelerates across its cradle clearance at 688 rad/s^2, arriving at 36–231 deg/s depending on fit—18 to 115 times the comparator. Whether that impact has damped out by release depends on a restitution model this study does not have, so tip-off is stated as a requirement on a cradle preload exceeding the 85 N couple reaction and on a release residual below about 1 N, against a release mechanism that remains undefined (Fig. 13). Recoil per shot is the satellite's momentum only, **64.1 N·s**, the sled's share returning through the brake. The solid model closes dry mass at **126.6 kg** (loaded 174.6 kg, centre of gravity 0.46 m from the breech), with the ironless stator at 7.3 kg of copper and formers; the system uses roughly a third of an ESPA Grande *mass* allocation. It does not currently fit the corresponding envelope: the closed CAD assembly spans 1839 mm along the track against the roughly 1270 mm of that class, because the arrest brake sits beyond the release point and the enclosure must span it. The third of those routes is the one taken: the target host class is a restartable upper stage, kick stage, or hosted orbital platform, and ESPA-Grande port compliance is not a requirement of this design. That is a re-scope, not a fix, and its cost is stated rather than absorbed. Shortening the track to 1270 mm would leave a 731 mm acceleration zone and, since velocity goes as $\sqrt{L}$, would cut ejection from 16.0 to 12.0 m/s—a 25% loss in the quantity every performance claim here rests on. The re-scope avoids that and gives up the ESPA-Grande port population in exchange. It also leaves the packaging question unanswerable for now rather than answered: no accommodation envelope for the target host class is public, so the 1839 mm length cannot presently be shown to fit anything, and obtaining that envelope is part of the same single data exchange Sec. VI identifies. This is an open item, and the honest description of it is that a measured failure against a published limit has become an unmeasured unknown against an undisclosed one.

VI. Generalized Host Integration

A. Interface Requirements for Any Host

VOLLEY-A, the attached mode analyzed here, requires four things of a host: a secondary-payload-class mechanical mount; 150–300 W of recharge power or an equivalent battery allocation; an attitude reference and enough control authority to null per-shot torque; and a disposal path consistent with debris rules. Nothing in the design binds it to one vehicle. Two

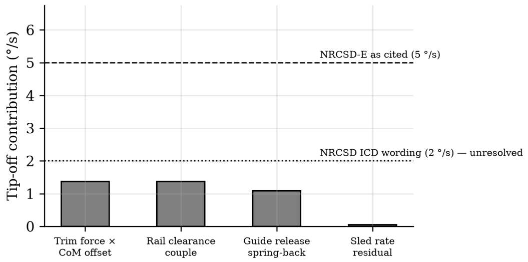

Fig. 13. Tip-off error budget against deployer requirement classes.

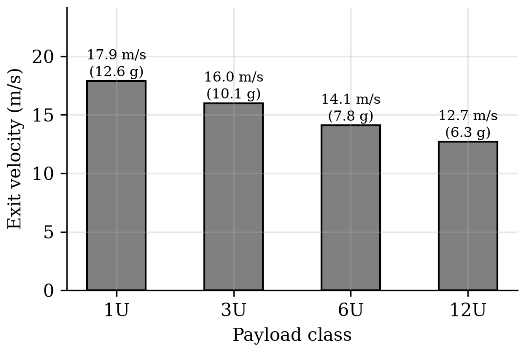

Fig. 14. Payload family: force-limited above 1U; velocity, not safety, is traded. †Counts are *volumetric*, calibrated so the 3U case returns the twelve the machine is laid out for. A subsequent fixed-cell design study replaces them where a cell geometry actually exists: 1U gives **36 per load at 2.125 kg**, not 40 at 1.92, because three 100 mm units plus two dividers leave 37.5 mm of a 340.5 mm cell unusable, and **ThinSat and 12U are refused outright** because two of their dimensions exceed the 100 mm cell section. The binding constraint is the 166 mm cassette width.

recoil facts generalize across hosts. First, correction propellant depends only on total ejected momentum, 0.787 kN·s for a full manifest, about 0.37 kg of hydrazine-class propellant regardless of host mass; host mass sets the per-shot rate disturbance, not the fuel bill. Second, the per-shot translation scales inversely with host mass (Table IV) and, for restartable stages, folds into disposal targeting. VOLLEY-F, a free-flyer variant with its own GNC, is the growth path for multi-day firing geometry and is not analyzed further here.

B. Candidate Hosts

Any restartable kick stage, extended-mission upper stage, or hosted orbital platform satisfies the interface. Two Indian candidates are worked as examples because both exist today. ISRO's POEM is the flown precedent: a spent PS4 operated as a three-axis-stabilized hosted platform with solar power, NavIC navigation, and helium attitude thrusters, retired by controlled reentry [14], [15]; it supplies everything VOLLEY-A borrows, and its zero-debris closeout is the regulatory template. Skyroot Aerospace's Vikram-1 carries a restartable liquid Orbit Adjustment Module (one Raman-2 engine, four Raman Mini thrusters, eight cold-gas thrusters, stage-tested through more than a thousand pulses [18]) whose stated multi-orbit deployment role is functionally the PS4's; against the

TABLE IV
PER-SHOT RECOIL (64.1 N·S) VERSUS HOST CLASS

Host class	ΔV per shot	12-shot total
200 kg free-flyer	328 mm/s	3.93 m/s
300 kg kick stage	219 mm/s	2.62 m/s
600 kg kick stage	109 mm/s	1.31 m/s
900 kg (PS4 class)	72.8 mm/s	0.87 m/s
420 t (station class)	0.2 mm/s	2.3 mm/s

vehicle's published 350 kg LEO capacity [19], a loaded VOLLEY is 34%, falling to 22% and 13% on the announced 550 kg and 900 kg family members, so early flights are dedicated demonstrations and later ones ordinary manifest items. One integration quantity cannot be closed from public data: the OAM's mass and control authority are undisclosed [18], which is why Table IV is parametric; obtaining stage mass, thruster impulse budget, and coast duration is the single data exchange that converts the analysis from parametric to specific for any candidate vehicle, Indian or otherwise. Comparable hosts elsewhere include kick stages evolved into mission platforms and ESPA-hosted propulsive rings; the interface asks the same four questions of each.

C. Integration Interfaces

The four-item interface resolves into concrete provisions. Mechanically the deployer bolts to a standard ESPA or ESPA-Grande ring through a canted bracket that passes the thrust line near the host centre of mass; the interface control document fixes the bolt pattern, envelope, and the published magnetic keep-out radius. Electrically it draws a 150–300 W recharge feed and a switched safety-inhibit line, with an isolated 28 V or bus-compatible tap and a dedicated ground per launch-vehicle EMC practice. The command interface is a serial link over which the host passes state vector and attitude-valid flags and the deployer returns telemetry: bank voltage, per-slot seat status, sled position, and shot confirmation. Sequencing is autonomous within a host-authorized window, the deployer computing each firing time from the supplied state and requesting only an attitude hold and the recoil-null authority quantified in Table IV; fault handling hands control back to the host safe-state on any inhibit trip. This division, deployer owning the firing logic and the host owning attitude and disposal, is what lets the same unit integrate on any compliant stage with only the electrical and command taps re-qualified per vehicle. It also admits a concept of operations in which the host is an active participant rather than a mounting surface. A spent upper stage is deorbited or left to decay regardless; operating it as a stabilized platform after primary separation, as POEM already demonstrates, converts that interval into deliveries. The stage repositions between altitude shells on its own reaction control, the deployer fires a subset of its manifest at each station at individually commanded velocities, and the stage performs its disposal burn when the magazine is empty. The envelope this reaches is bounded rather than open. A 50 km shell change costs 27.8 m/s as a two-burn

TABLE V
SERVICE TIERS FOR SMALL-SATELLITE ORBIT PLACEMENT

Attribute	Spring deployer	VOLLEY (this work)	Transfer vehicl
Velocity authority	∼2 m/s fixed	2–20 m/s programmable	100s m/s
Per-satellite targeting	no	yes, ±0.027 m/s	yes
Satellite modification	none	none	rides as payloa
System mass, 12×3U	10–20 kg	125 kg loaded	100s kg + prop
Plane/LTAN change	no	no (≤0.2°)	yes
Flight status	flown, extensive	design study	announced

transfer at these altitudes, so a 100 m/s host budget reaches four shells and delivers a fleet spanning 269 km of altitude; the along-track distribution is the deployer's own and costs the host nothing; and nodal separation develops for free from differential J_2, reaching 45° over a 90-day campaign at that budget and 13° with no host manoeuvring at all. Plane change is excluded: at 133 m/s per degree it exceeds the entire shot by a factor of eight and would consume a stage budget without repositioning it. The division of labour is worth stating precisely, because it is not flattering in one direction: above roughly 100 m/s of host budget the *stage* supplies most of the altitude range, and the deployer supplies the distribution within it. Neither substitutes for the other, and the altitude half of any multi-orbit claim is bought with host propellant. The interface has a second direction that the four items above do not cover. Each of them, and the published keep-out radius, states what the deployer asks of a *host*; none states what the deployer does to the *satellite it carries*. The magnetic environment inside the magazine is the case where that distinction has consequences, and it is specified in Sec. XII: the payload envelope sits at 611× a representative magnetometer full scale at its near face and 3.4× at its far face, continuously rather than only during a shot, because the array is a permanent magnet. The claim that the customer satellite is never modified holds mechanically and electrically—no armature, plating, harness, or separation system is imposed—but it is not established magnetically, and a payload environment specification rather than a host keep-out is what an integrator needs in order to check compatibility.

D. Position in the Deployment Landscape

Deployment services today form two tiers: flown low-velocity springs (EXOpod-class hardware near 2 m/s with hundreds of deployments [20]) and announced propulsive transfer vehicles offering hundreds of m/s to whole buses, none yet flown in the small-OTV class relevant here [26]. VOLLEY does not compete with either on its own terms: it cannot change a plane, and a spring cannot place twelve satellites on twelve chosen ellipses. Table V states the comparison. The middle tier, per-satellite deterministic velocity at deployer-class mass serving the propulsion-less payload population, currently has no demonstrated occupant.

E. Demonstration Path

F. What Air Costs a Ground Test

The machine operates in vacuum and every velocity above is computed with no aerodynamic term, but the full-scale

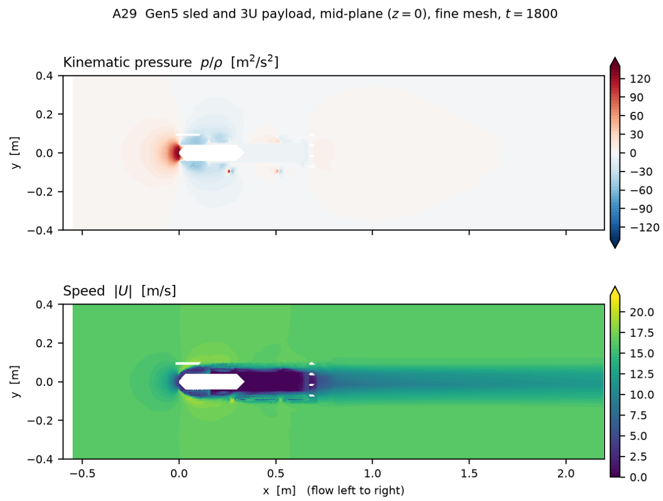

Fig. 15. Mid-plane slice ($z = 0$) of the converged fine-mesh solution: kinematic pressure above, speed below, flow left to right. Stagnation on the forward face, separation at the shoulders, and a wake that has not recovered by $x = 2.2$ m. The pressure term shown here is integrated directly from the solved field; **the viscous term is not solved** and is bounded by a turbulent flat-plate correlation over the meshed wetted area, as the text states.

demonstration fires a mass simulator down the track in a laboratory at sea level, so the comparison that test exists to make is between a measured velocity and a computed one that omits the air. A steady RANS solution (k-ω SST) around the sled and payload *as generated*, meshed directly from the CAD surfaces rather than from an idealised body, gives a drag coefficient of **0.52** on a 201.6 cm^2 projected frontal area at $Re = 7.0 \times 10^5$, i.e. 1.73 N at exit velocity. Because the profile is position-scheduled, v^2 rises linearly with distance and so does the drag, making the work over the 1.30 m acceleration zone 1.13 J and the exit-velocity deficit **5.1 mm/s**, or 0.031 % (Fig. 17).

That figure is small against the design point and *not* small against the quantity a ground test resolves: it is **19 % of the 3σ closed-loop dispersion**. A full-scale test therefore needs no vacuum chamber, but every velocity it reports must carry the air correction, or it is compared to the model with an error one-fifth the size of the dispersion being verified. The distinction depends on how the test is run: open-loop, the deficit appears as 5.1 mm/s of velocity; closed-loop, the servo nulls it and it appears instead as 1.73 N of additional commanded force, 0.125 % of the shot.

Two properties of the solution are reported rather than presented as convergence. The velocity residuals fall three orders and then plateau near 5×10^{-2}, which is what a steady solver does on a massively separated wake, so the force is a mean over a window of iterations quoted with its spread, 1.734 ± 0.144 N. And the viscous contribution is a turbulent flat-plate bound over the meshed wetted area rather than a solved wall shear stress, 17.7 % of the total; the pressure term, which dominates, is integrated directly from the solved field over the body surface. Adding the stator plates *reduces* drag

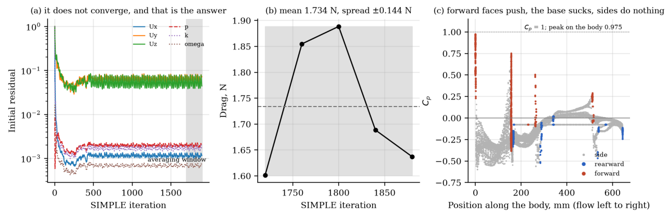

Fig. 16. Force history and mesh convergence for the same case. The force is a mean over a window of iterations quoted with its spread, 1.734 ± 0.144 N, because a steady solver on a massively separated wake plateaus rather than converges; $6.3\times$ the cells moves the answer by 4.86 %.

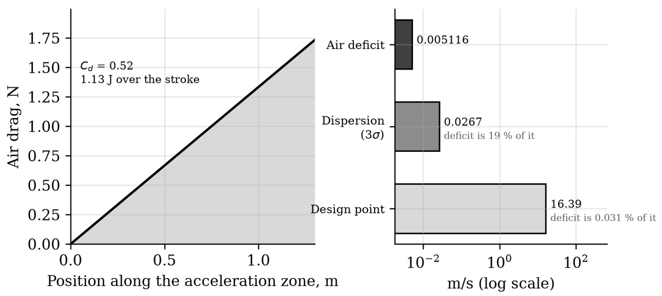

Fig. 17. Air drag along the acceleration zone, and the resulting exit-velocity deficit set against the design point and against the dispersion a ground test exists to resolve.

by 13 % and halves the oscillation, the plate acting as a wake splitter rather than as a confining wall, so the free-stream figure quoted above is the conservative one.

The programme is ground-testable to an unusual degree: a full-scale 1.5 m track firing a mass simulator into the eddy brake, with the complete feed cycle, sits within a university laboratory budget and converts the velocity, dispersion, and tip-off claims into measured quantities. The flight step is a captivecarry demonstration, full sled cycles with a payload never released, requiring no deployment approvals while gathering the reaction-load and servo data that certify the firing event; any hosted platform of the POEM class serves.

VII. Sensitivity and Uncertainty

Because several inputs carry manufacturing or environmental spread, the rated point is bracketed by a first-order sensitivity analysis (Table VI). Each coefficient is the local partial derivative of exit velocity or thrust with respect to the listed variable, evaluated at the rated point. The dominant contributors are payload mass and magnet remanence; the velocity model is mild in all of them, and the closed-loop servo further collapses the delivered dispersion to 0.0274 m/s (Sec. V-A) because it commands current against measured position rather than trusting the plant. Root-sum-square combination of the manufacturing terms (magnet grade $\pm 2\%$, gap ± 0.05 mm, resistance $\pm 5\%$, ESR $\pm 20\%$) gives an open-loop velocity spread of $\pm 2.1\%$, consistent with the Monte Carlo of Fig. 8. Environmental terms are handled by rating margin: the N45SH remanence temperature coefficient of -0.11%/K over a ± 40 K operating band moves the thrust constant by $\mp 4.4\%$, so the worst-case cold-to-hot requirement of 132 kA/m stays inside the 140 kA/m current rating without re-tuning. Atmospheric and solar-activity spread do not affect the machine and enter only through orbital lifetime, where Sec. V-B records that the sweep previously offered as evidence of invariance could not, by construction, have detected a change.

TABLE VI
First-order Sensitivities at the Rated Point

Variable	Affects	Sensitivity
Payload mass	v	$-\frac{1}{2}$ (%/% of m_{tot})
Magnet remanence B_r	F, v	$+1.0,\ +0.5$ %/%
Current density	F, v	$+1.0,\ +0.5$ %/%
Winding resistance	η	-0.15 %/%
Capacitor ESR	sag	$+0.04$ V/mΩ
Gap asymmetry	K_t	-13 %/mm
Sensor noise (8 mm/s)	dispersion	sets 3σ floor
Temperature (B_r)	K_t	-0.11 %/K

VIII. Mechanical Analysis

The structural load path is sized to the two governing cases: launch vibration during ascent and the arrest impulse during operation. Fastener and pin margins are quoted on ultimate strength at a 1.4 design factor per common launch-hardware practice, consistent with the qualification philosophy of GEVS [21] and ECSS structural standards [22].

Track stiffness. The dominant flight requirement is a first natural frequency above the launch-vehicle primary band, conventionally 70–100 Hz for secondary structure. Modeling the twin-longeron track as a uniform beam carrying the 20 kg stator distributed mass, the fundamental is $f_1 = (\lambda_1^2/2\pi L^2)\sqrt{EI/\mu}$ with EI from two 7075 box sections. A simply-supported idealization gives 48 Hz, but bracketing both ends into the ESPA ring (fixed-fixed) raises the first mode to 109 Hz, clearing the requirement; the design therefore specifies end-fixed mounting with an intermediate rib.

Inter-array attraction. The two Halbach faces attract across the gap with a Maxwell stress $p = B^2/2\mu_0 \approx 120$ kPa at the 0.55 T mean working field, or 3.7 kN over the array footprint. Carried as a bending load into the titanium side plates, this produces 33 MPa against the 880 MPa Ti-6Al-4V yield, a margin above 20; the attraction is structurally minor and, being static, does not fatigue-cycle. The flat-plate expression overstates the force: a three-dimensional integration of the field gradient over the meshed blocks returns 2.69 kN, some 37% lower, because the assumed 0.55 T is not the mean normal field over the separation plane. Jensen's inequality acts in the opposite direction and does not explain the overestimate. Extending the Maxwell-stress plane beyond the magnet footprint gives 2.680 kN, within 0.25% of the 2.687 kN volume-force calculation; this is model-to-model agreement, not experimental validation. The earlier finite-element load remains conservative.

Arrest and roller loads. At the 200 g arrest cap the 9.445 kg sled develops 18.5 kN axially, reacted by the ring-spring stop; the pitch couple during braking loads the guide rollers to roughly 1.48 kN per pair. This is the load path most affected by adopting the CAD-derived sled mass—it very nearly doubles—and it is the one place where the heavier sled makes the machine harder rather than merely slower. Bearing selection at 1.48 kN per pair is within the rating of a 440C/Si_3N_4 hybrid of this size, but the reuse duty is not addressed here: the sled is braked twelve times per campaign in vacuum, and no lubrication or contact-life assessment has been performed. The abort latch, engaged only for a commanded abort, carries the 13.4 kg carriage at the abort deceleration, 0.42 kN, through dual A-286 pins at a margin above eight.

Fatigue and tolerance. Operational cycling is benign: the magnet bond sees 0.12 MPa per shot against a 10 MPa allowable, far below any endurance concern over the few hundred shots of a service life. Positional accuracy is set by the winding-to-magnet gap, whose -13%/mm thrust sensitivity dictates a ± 0.05 mm shim tolerance to hold thrust within $\pm 0.65\%$; this is a standard precision-shim assembly, not a novel requirement.

A. Architecture reliability, and the bar it sets

A spring dispenser is twelve independent one-shot mechanisms in parallel: one failure costs one satellite. This design is one mechanism in series with itself, cycled twelve times. The sled, stator, converter, energy store, sequencer and brake serve every shot, and the escapement and retention gate cycle twelve times each, so a failure at shot k forfeits shots k through twelve. Enumerating the shot chain, **nine of thirteen elements forfeit the remaining manifest on a single failure, against zero for a spring dispenser**, and nine shared elements over twelve cycles is 108 opportunities to fail. This is structural and no component quality removes it.

What component quality does change is where the comparison lands. Writing p for the probability that the shared chain survives one shot, expected delivery is $\sum_{k=1}^{12} p^k$, which departs from a spring's independent $12q$ quickly. Against a spring at $q = 0.99$, matching on *satellite count* requires $p = 0.9985$, which is not a realistic target for a twelve-cycle electromechanical system without flight heritage. Matching on *delivered orbital life* requires only $p = 0.9347$, because each delivered satellite is worth 1.50 times a spring-deployed one. Expressed per element per cycle, the requirement is $r \geq 0.99326$.

That is the first quantitative reliability requirement this design has carried, and it is not yet met or measured: no cycle-life test exists for the escapement, the gate or the sled. Two mitigations are real and are deliberately not credited above, because crediting them without analysis is the optimism the exercise exists to avoid. Retention gates are per cassette, so one gate fault forfeits six rather than twelve. And the winding is segmented: a dead segment is a length the sled coasts over rather than a stopped machine, so at four segments a failure downstream of the breech still exits at 14.19 m/s, 86.6% of nominal. The exception is the breech segment itself, where no force acts on a stationary sled and the shot never starts, which makes duplicating or overlapping that one segment the cheapest available reliability action.

IX. Thermal Analysis

Thermally the deployer is a pulsed, low-duty machine: each shot deposits heat in milliseconds, and the following minutes-to-orbit reject it. The per-shot budget is 850 J in the stator copper (835 J over the shot, 15 J over the regenerative braking pulse), 935 J in the brake fin, 111 J in the converter, and 91 J in the supercapacitor equivalent series resistance. Over a full twelve-shot campaign the total is 24.3 kJ; deposited into the roughly 13.5 kJ/K thermal mass of the structure this is a 1.8 K bulk rise, so no shot ever sees a meaningfully preheated machine and the adiabatic per-shot rises quoted earlier (0.35 K coil, 7.1 K fin) are the design drivers rather than steady state. The brake fin is the hottest element. Its corrected 0.344 kg CAD mass gives a 7.1 K adiabatic rise per shot and an 85 K twelve-shot no-cooling bound. No transient conduction or radiation model establishes how much of that rise decays between shots, and the repository carries both 10–20 s and 1200 s cadence assumptions. Recovering a quarter of the sled's energy is worth more thermally than electrically: it is the brake fin, not the bank, that the reduction relieves. The existing 0.32 m^2 radiator figure follows from an assumed 130 W rejection case, not from the cadence model. A transient thermal network and a resolved campaign schedule are required before either the between-shot fin temperature or radiator sufficiency is established. The orbital environment sets the endpoints: a hot-case beta-angle attitude with full solar and albedo input, and a cold-case eclipse, bound the supercapacitor and converter baseplates, both of which are specified to survive -40 to $+60$°C with survival heaters covering the cold case at a few watts. The N45SH grade is selected precisely so the magnets tolerate the hot case without irreversible loss.

X. Power Electronics

The drive is a three-phase two-level voltage-source inverter of six SiC MOSFET half-bridges (1200 V devices on the 96 V rail, a 60% derating that accommodates switching overshoot), commutated by field-oriented control synchronized to the sled position encoder. Isolated gate drivers with desaturation detection and active Miller clamp switch at 20–40 kHz, low enough to keep switching loss within the converter's 97 J/shot budget. The winding inductance that filters the resulting ripple is computed rather than assumed, which it previously was not. A sheet-current model is turns-invariant, so the turns count is fixed instead by requiring the inverter to synthesize the phase voltage the machine demands at rated speed out of the sagged 90.9 V bus: this gives 19.7 μH, a 373 A peak phase current—distinct from, and larger than, the 320 A drawn at the DC link—and 25.2 mΩ per phase at unity modulation index at exit. Peak-to-peak ripple is then 16% of peak at 20 kHz and 8% at 40 kHz, adding 3.7 J to an 835 J copper budget. The loss consequence is therefore negligible across the range, but only its upper end produces a current that can fairly be described as filtered. The two-dimensional energy

TABLE VII
PRINCIPAL FAILURE MODES AND MITIGATIONS

Failure	Effect	Mitigation
Sled jam	**Manifest stranded**	Caged escapement; controlled-disposal host. **The dual-cassette split bounds an *escapement or gate* fault to half the manifest, but not a sled fault: one sled serves all twelve cells**
Latch fails	Early release	Latch loaded only on abort; inertia retains during accel; three-inhibit fire chain
Sensor fault	Wrong velocity	Redundant seat switches, encoder, photogate cross-check; no-fire on disagreement
Capacitor cell	Reduced energy	32 series cells; one shorted cell drops rail 3%, within margin
Inverter device	Phase loss	Segmented winding; dual gate drive; degraded-thrust continue
Magnet debond	Field loss	$84\times$ bond margin at 200 g; taper-limited arrest
CubeSat misalign	Feed jam / tip-off	Seat switches + optical check; no-fire interlock
Brake fault	Sled overrun	Ring-spring hard stop backs eddy brake; motor regen trim
Controller fault	Loss of shot	Safe-state inhibits; watchdog; abort to 45% stroke

method omits end turns, so the inductance is a lower bound and the ripple figures upper bounds. Protection comprises cycle-by-cycle current limiting in the gate-drive logic, DC-bus overvoltage clamping to absorb regenerative transients, and desaturation shutdown on device fault. The 96 V bank is thirty-two series cells; passive balancing resistors equalize cell voltage over the long inter-shot interval, adequate because the duty cycle leaves ample balancing time, and a bleed network plus commanded crowbar passivate the bank at end of life. Electromagnetic interference from the switching edges is contained by keeping the high-di/dt loop area small, filtering the bank input, and enclosing the converter in a shielded housing; the firing transient is scheduled after primary separation so no adjacent live payload is exposed. Redundancy is applied where a single fault would strand the manifest: the winding is segmented for fault isolation, so a shorted coil degrades thrust rather than ending the campaign; the segments are driven as a single energised section rather than block-commutated, which costs efficiency and is stated explicitly because a segmented long-stator machine would normally imply the latter. The gate-drive and sensing chains are dual.

XI. RELIABILITY AND FAILURE MODES

Table VII summarizes the principal failure modes in an FMEA form, with the mitigation designed into the architecture. The design philosophy throughout is that the satellite is retained passively during ascent, that no single electrical fault strands the manifest, and that any credible mechanism fault degrades gracefully rather than releasing a satellite in an uncontrolled state.

XII. SPACE ENVIRONMENT, EMC, AND STANDARDS

The materials selection of Sec. III-F is driven by the low-Earth-orbit environment. Outgassing is controlled to ASTM E595 limits (TML $< 1\%$, CVCM $< 0.1\%$) for every polymer, with dry-film MoS_2 and confined PFPE grease replacing wet lubrication that would cold-weld or migrate in vacuum [22]. Atomic-oxygen exposure is handled by hard-anodized aluminium and SiO_2-overcoated multilayer insulation on ram surfaces. Thermal cycling and radiation are within the ratings of the selected sinter, cells, and SiC devices for a mission of this duration; permanent-magnet aging over a few-hundred-shot life is negligible given the operating field sits well below the knee. Vibration and shock survivability are demonstrated by the 109 Hz first mode and the launch-load margins of Sec. VI.

Electromagnetic compatibility centres on the stray field of the Halbach sled. The verified back-face profile of 22.7 mT at 10 mm, 4.3 mT at 20 mm, and 0.4 mT at 50 mm sets a magnetic keep-out: stored satellites are parked outside the near field by the cassette geometry, and thin silicon-steel septa on the cassette inner walls attenuate the static field further. The self-shielding weak side is oriented outward by design, and a keep-out radius is published in the interface control document with a waiver process for magnetically sensitive payloads. That keep-out governs the field *outside* the deployer, and it should not be read as bounding the environment *inside* it, which is far more severe: profiling the array field across the 3U payload envelope gives 61.1 mT at the near face and 0.341 mT at the far face, crossing below a representative ± 100 μT attitude-magnetometer full scale only at 251 mm and below Earth's own field at 332 mm. Since the envelope ends at 120 mm, every part of a cradled payload sits above both, from $611\times$ full scale at the near face to $3.4\times$ at the far face, and the static magnet field rather than the switching transient is the dominant term. The operational consequences—a magnetometer that reads saturated and a magnetorquer that cannot be commanded until after ejection—are recoverable; permanent magnetisation of soft-magnetic payload structure would not be, and bounding it requires a payload materials list that this study does not have, so it is reported as unquantified rather than negligible. The switching transient is contained as in Sec. X and fired only after separation, so induced currents in adjacent radios or instruments are avoided by timing as well as shielding.

XIII. MANUFACTURING, COST, AND SCALABILITY

The machine is deliberately conventional to build. The stator is an ironless copper winding wound on a machined PEEK former to a ± 0.1 mm slot tolerance and vacuum-impregnated with E595-qualified epoxy; the sled magnets are individually plated N45SH blocks bonded into a titanium carrier in a magnetized-assembly fixture that reacts the inter-block forces, then shimmed to the ± 0.05 mm gap. Assembly proceeds stator-to-track, track-to-structure, cassette population last so the satellites are installed only after the mechanism is verified, mirroring the quad-pack integration practice of heritage deployers. Calibration is a ground exercise: each unit fires a mass simulator across an instrumented catch to map currenttovelocity before flight. Because the sled and track are reusable, maintenance between campaigns is inspection rather than refurbishment.

On cost, a parametric estimate rather than a business case, and one that corrects an earlier assumption in this work. The recurring hardware was previously described as dominated by the magnet set and the SiC drive. A bill of materials built against the mass model does not support that: the magnet set is roughly 5% of recurring hardware cost and the drive 13%, while avionics, the supercapacitor bank and the position-sensing chain together account for approximately half. The ordering is robust even though the prices are not—no line item priced here carries a vendor quotation, but doubling the magnet cost still leaves it below a tenth of the total. The practical consequence is that this machine is an avionics and energy-storage cost problem rather than a magnetics one, which is the opposite of where the design effort has gone. The per-shot energy cost is the striking figure. At 2.78 kJ drawn per ejection the electrical energy is under one watt-hour, met by solar recharge between shots at zero consumable cost, against the propellant a transfer vehicle would expend for the same manoeuvre. A spring deployer has no energy cost but also no programmability; VOLLEY buys deterministic per-satellite velocity for less than a watt-hour a shot.

Scalability follows the governing equations. Larger CubeSats are already covered by the payload family of Table III, trading velocity for mass at fixed force; higher velocity comes from a longer track ($v \propto \sqrt{L}$) or higher current rating; more satellites come from deeper cassettes at a linear mass cost. The architecture extends to lunar or deep-space secondary deployment unchanged in principle, since it needs no atmosphere and no propellant, with only the thermal and radiation ratings revisited for the destination environment.

XIV. Operational Concept and Maturation

A mission proceeds through launch with the deployer dormant and the sled cam-locked; commissioning and health check once the host reaches orbit; bank charge from the host or internal battery; and a deployment campaign in which the sequencer computes each firing time from the host state, verifies the three-inhibit chain, fires, and confirms release by photogate before advancing the magazine. Contingencies branch to abort-and-restow within the first 45% of any stroke, or to safe the bank and hold. The campaign closes with bank passivation and, for a hosted stage, folding the accumulated recoil into the host's controlled-disposal burn, following the zero-debris precedent of POEM-3 [14].

The maturation path is unusually testable on the ground. The present work is the analytical-model phase, TRL 2–3, with the independent field verification of Sec. IV-B and the cross-validated astrodynamics of Sec. IV-C. A benchtop single-cassette track at reduced energy advances to TRL 4; a full 1.5 m track firing a mass simulator into the eddy brake, with the complete feed cycle, reaches TRL 5 and converts the velocity, dispersion, and tip-off claims into measured data within a university laboratory budget; vacuum and vibration qualification of the mechanism follows at TRL 6. The flight step is the captivecarry demonstration of Sec. VI-D, firing full cycles with a payload never released, reaching TRL 7 without any deployment approval, before an operational orbital deployment.

XV. Limitations

Six limits bound the claims. The atmosphere model is static, so absolute lifetimes carry severalfold uncertainty; an earlier version of this paper offered the invariance of the lifetime *ratio* as the defensible result in their place, and Sec. V-B records why that was wrong—the sweep supporting it could not have detected a change. The ratio is now quoted at a stated activity level and carries the same uncertainty as everything else derived from a static atmosphere. The thrust constant is winding-resolved in two dimensions; three-dimensional end effects of a few percent remain for finite-element closure. The brake law is first-order plate drag, acceptable at an order-of-magnitude authority margin. Conjunction figures, although refined at candidate minima, do not replace per-shot collision-avoidance products. Host budgets are parametric wherever stage properties are not public. Masses now derive from detailed CAD rather than the parametric solid model used earlier, and the two disagree: exact solid volumes give a 9.445 kg sled against the 4.86 kg the parametric model assumed. The CAD-derived value is adopted here and sets the headline velocity, but it is the as-drawn, unpocketed geometry—the structural analysis reports a 17-fold stress margin on the chassis plate, so a rib-stiffened redesign would recover mass, and none has been evaluated. The sensitivity, mechanical, thermal, power-electronics, and reliability analyses added here (Secs. VII–XIV) address the principal engineering questions only at analysis level. Several remain feasibility-significant. The commercial capacitor strings examined do not meet the ESR requirement; a flywheel motor-generator has since been shown to clear it at 35 mΩ against the 68 mΩ ceiling, needing no architecture change, but at mass parity rather than saving. Three-dimensional field closure is now partly resolved and the result is unfavourable: resolving the field across the array's 90 mm depth, rather than sampling it at the centre plane and multiplying, reduces K_t by **4.42%** to 10.54 N per kA/m and v_{exit} to 16.03 m/s. **That correction is computed and deliberately not applied here**, because it has not yet been checked by a method that solves a field equation in three dimensions, and re-baselining onto an unchecked number would repeat the error it identifies. And hardware validation has not begun.

XVI. Conclusion

A magazine-fed linear-motor deployer addresses, on numerically cross-checked model outputs, a deployment regime that current hardware leaves empty. The design ejects unmodified 3U CubeSats at 16.0 m/s within standard qualification loads, at 18.8% electricaltopayload efficiency net of regenerative recovery and 0.0274 m/s dispersion, from a 126.6 kg system compatible with any restartable stage or hosted platform meeting a four-item interface. That authority converts into utility no spring provides and no transfer vehicle economically matches for this payload class: a 1.60-fold lifetime multiplier per shot at mean solar activity, from a 28.8 km rise in semi-major axis that neither a spring, nor differential drag, nor the release schedule can produce. The velocity is 6.4 times the fastest published spring comparator and is set by a sled mass computed from detailed CAD solid volumes rather than estimated; recovering the 20.4 m/s of the earlier parametric design is a mass-reduction problem with a quantified target, not an open question. The claim is deliberately narrow, and external review has narrowed it further: the linear motor is selected for *commandability* rather than performance, since a screw cannot reach the velocity at all and a $\sim$1.8 kg staged spring can, failing only in that its velocity is built in rather than commanded. What this machine sells is not a distribution schedule—in-track phase is available for the cost of waiting between releases, and an earlier version of this paper compared it against differential drag rather than against a clock—but an orbit the host is not in, and it is bought at a cost in mass, complexity and shared-failure exposure that is stated here rather than discovered later. The concept asks one question of any launch provider, the mass and control authority of its stage, and offers in return a way to sell every rideshare customer their own orbit.

XVII. Future work: a second architecture

The design analysed above is the one this paper defends. A second architecture has since been developed in the project record and is stated here as direction rather than result. It removes the launcher from the deployer: the payload is accelerated directly by cold gas along a rail that a spent upper stage already provides, with no mover, no stator, no pulse-power chain, no brake and no return stroke. The motive is the mass case reported in Sec. XV—deleting subsystems attacks the numerator that Sec. V-B cannot—and the cost is a dependence on a host vehicle remaining alive past passivation, which no launch provider has agreed to. Nothing in that architecture has been measured; its release mechanism does not exist; and its one electromagnetic element, a short trim stator that corrects the velocity the gas produces, has been suspended pending a seal-friction measurement that decides whether it is needed at all. It is therefore future work and not a result, and it is reported in the project repository rather than here, for the same reason the three-dimensional field correction in Sec. XV is computed and deliberately not applied: re-baselining onto unvalidated numbers would repeat the error this paper spends its limitations section identifying.

References

[1] C. Lewis, "Failure of the ball-lock mechanism on the NanoRacks CubeSat deployer," in *Proc. 44th Aerospace Mechanisms Symp.*, 2018.

[2] JAXA, "JEM payload accommodation handbook, vol. 8: Small satellite deployment interface control document," via UNOOSA KiboCUBE.

[3] NASA SSVI, "State-of-the-art of small spacecraft technology: Integration, launch, deployment, and orbital transport."

[4] "Design and analysis of a new deployer for the in orbit release of multiple stacked CubeSats," *Remote Sensing*, vol. 14, no. 17, 4205, 2022.

[5] B. N. Turman *et al.*, "Coilgun launcher for nanosatellites," Sandia National Laboratories, OSTI.

[6] R. J. Kaye *et al.*, "Electromagnetic coilgun launcher for space applications," Sandia National Laboratories, OSTI 125180.

[7] K. Halbach, "Design of permanent multipole magnets with oriented rare earth cobalt material," *Nucl. Instrum. Methods*, vol. 169, pp. 1–10, 1980.

[8] R. F. Post and D. D. Ryutov, "The Inductrack: A simpler approach to magnetic levitation," *IEEE Trans. Appl. Supercond.*, vol. 10, no. 1, pp. 901–904, 2000.

[9] F. Faure *et al.*, "Qualification of COTS supercapacitors for space applications," *ESA Space Passive Component Days*, 2018.

[10] "Towards supercapacitors in space applications," *European Space Power Conf.*, 2017.

[11] E. Diez-Jimenez, C. Alén-Cordero, R. Alcover-Sánchez, and E. Corral-Abad, "Modelling and test of an integrated magnetic spring-eddy current damper for space applications," *Actuators*, vol. 10, no. 1, art. 8, 2021; together with commercial eddy-current damper flight-heritage documentation, not independently retrieved.

[12] C. Foster *et al.*, "Constellation phasing with differential drag on Planet Labs satellites," *J. Spacecraft and Rockets*, vol. 55, no. 2, 2018.

[13] C. Foster, H. Hallam, and J. Mason, "Orbit determination and differential-drag control of Planet Labs CubeSat constellations," arXiv:1509.03270, 2015.

[14] ISRO, "PSLV-C58/XPoSat mission: POEM-3 accomplishes zero orbital debris mission," Mar. 2024.

[15] ISRO, "POEM-4 completes 1000 orbits in space," 2025.

[16] NanoRacks, "NanoRacks CubeSat deployer interface definition document," 2018.

[17] D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, 4th ed. Microcosm Press, 2013.

[18] Skyroot Aerospace, "Orbit Adjustment Module full stage-level test," public statement, Oct. 14, 2025.

[19] J. Foust, "Skyroot prepares for first orbital launch attempt," *SpaceNews*, Jul. 7, 2026.

[20] Exolaunch, "CarboNIX" and "EXOpod Nova" product documentation; Exolaunch–Skyroot partnership announcement, Oct. 14, 2025.

[21] NASA, "General Environmental Verification Standard (GEVS)," GSFC-STD-7000A, 2013.

[22] ECSS, "Space engineering: Structural and mechanical / materials standards," ECSS-E-ST-32 and ECSS-Q-ST-70 series.

[23] California Polytechnic State Univ., "CubeSat Design Specification," Rev. 14, 2020.

[24] E. Kulu, "Nanosats Database," nanosats.eu, counts as of Jan. 1, 2026.

[25] Novaspace, "Prospects for the Small Satellite Market," 11th ed., 2026. Market forecast, not a measurement.

[26] Bellatrix Aerospace, "Bellatrix Aerospace and NSIL sign MoU to integrate Bellatrix's OTV in NSIL's launch missions," Oct. 9, 2024.

[27] H. Feng, Y. Yang, and Y. Wu, "Design and reachable domain analysis of on-orbit electromagnetic launcher for CubeSats," *Int. J. Aerospace Eng.*, vol. 2025, art. 3000765, 2025.

[28] D. Xu, H. Yue, Y. Zhao, F. Yang, J. Wu, X. Pan, T. Tang, and Y. Zhang, "Improved A* algorithm for path planning based on CubeSats in-orbit electromagnetic transfer system," *Aerospace*, vol. 11, no. 5, art. 394, 2024.

[29] Y. Zhao, Y. Zhang, Z. Zhao, C. Li, L. Zhang, X. Yang, H. Yue, *et al.*, "Simulation analysis and experimental verification of the transport characteristics of a high-volume CubeSat storage device," *Aerospace*, vol. 12, no. 6, art. 466, 2025.

[30] Y. Zhao, H. Yue, X. Mu, X. Yang, and F. Yang, "Design and analysis of a new deployer for the in-orbit release of multiple stacked CubeSats," *Remote Sensing*, vol. 14, no. 17, art. 4205, 2022.

[31] M. Einat and Y. Orbach, "A multi-stage 130 m/s reluctance linear electromagnetic launcher," *Sci. Rep.*, vol. 13, 2023.